

SVM = Support Vector m

→ Supervised ml Algorithm

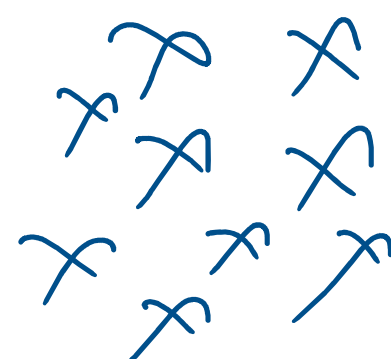
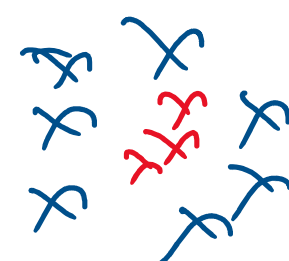
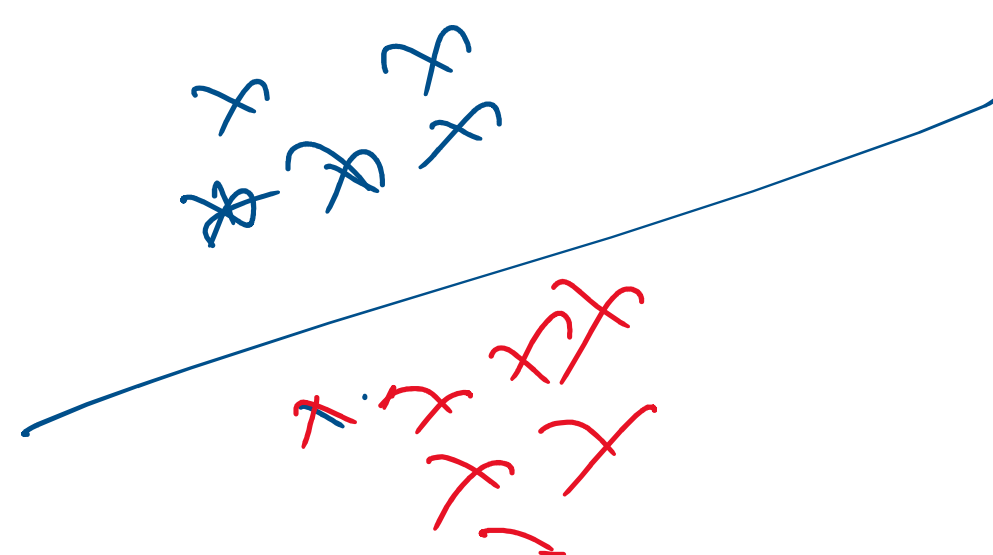
→ classification ☒ popular
Regression

Linear

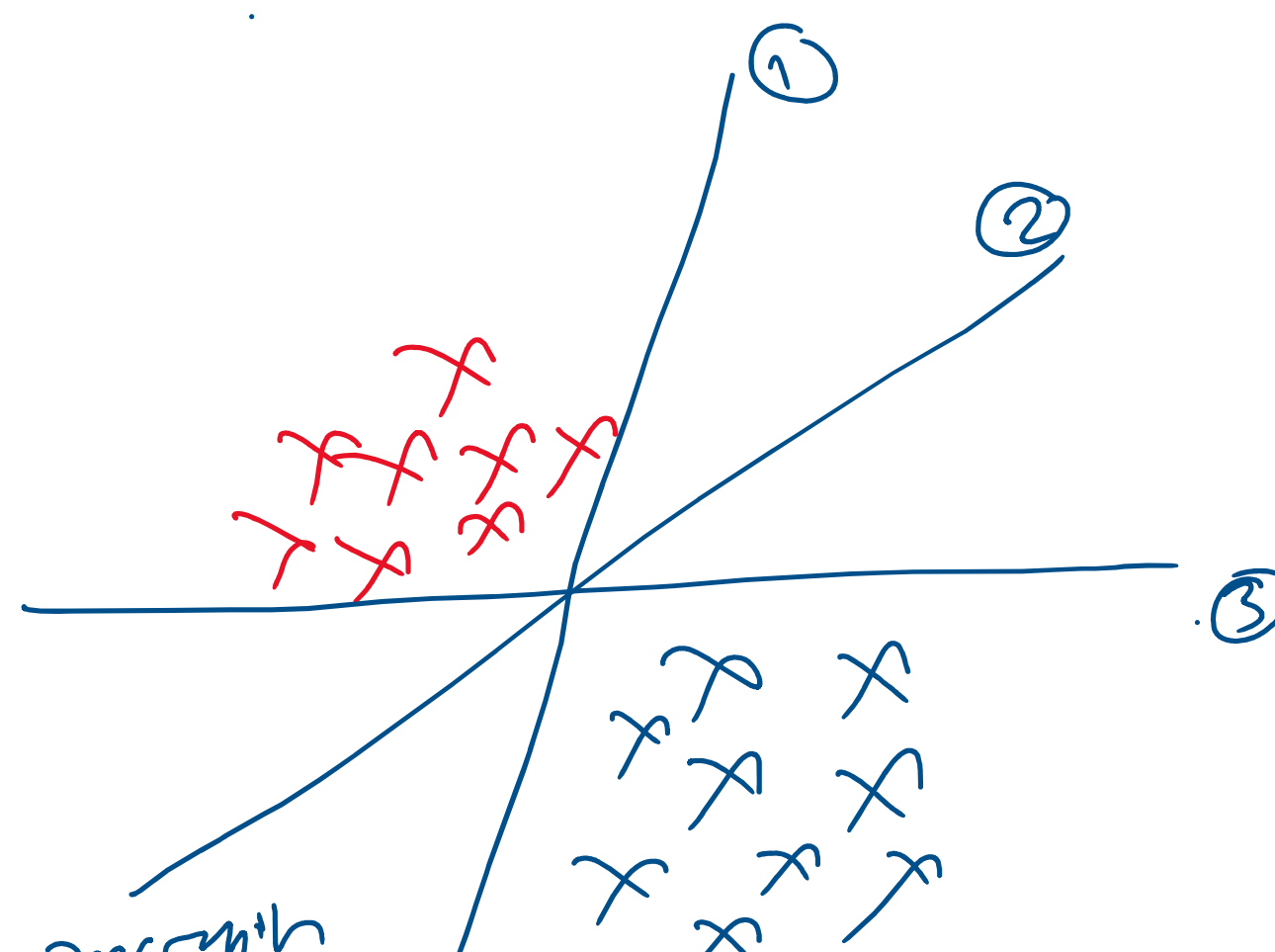
non-linear

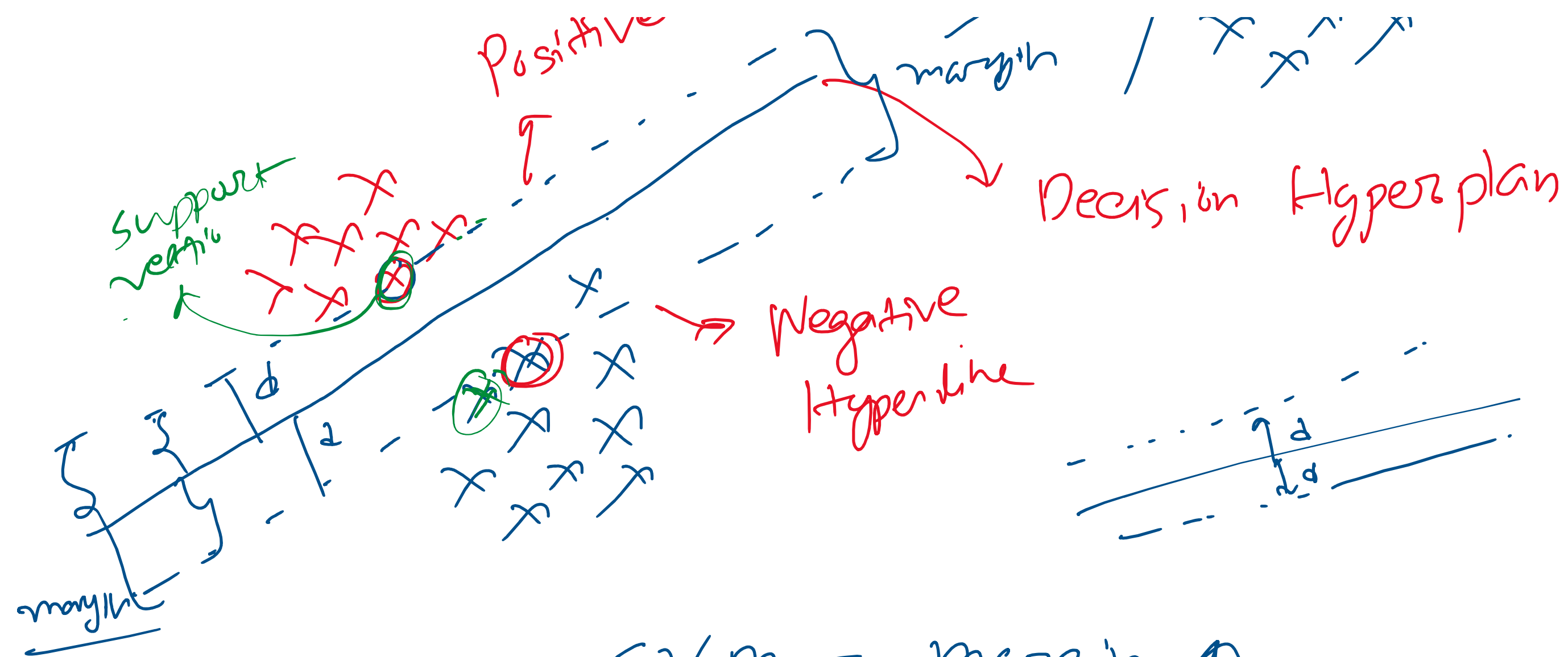
→ Feature Invariance
→ computation

Outlier

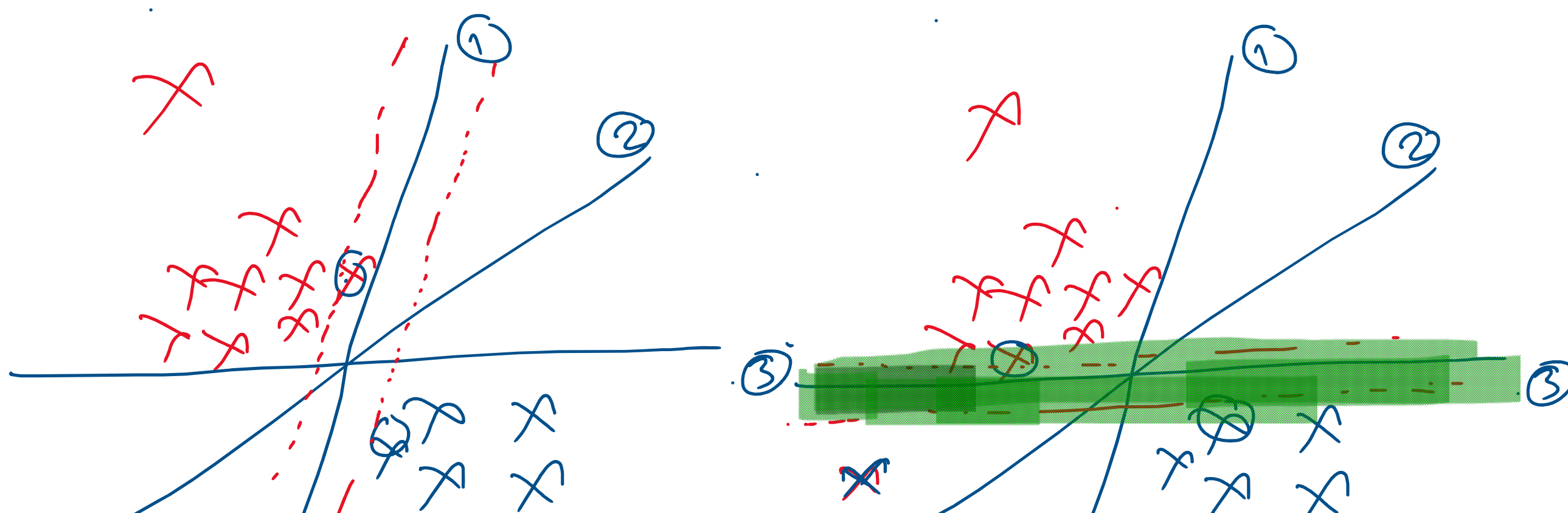
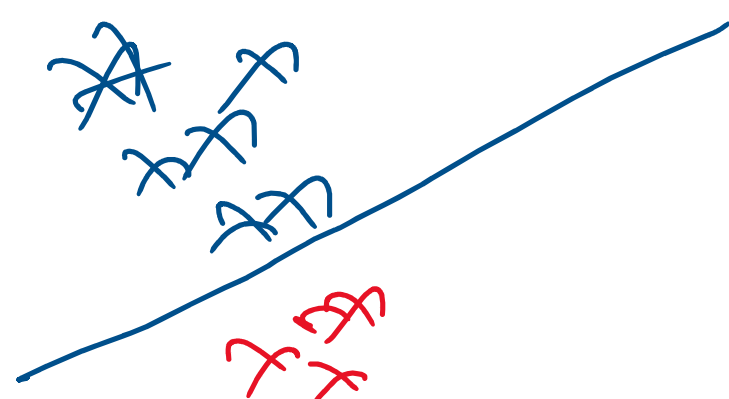


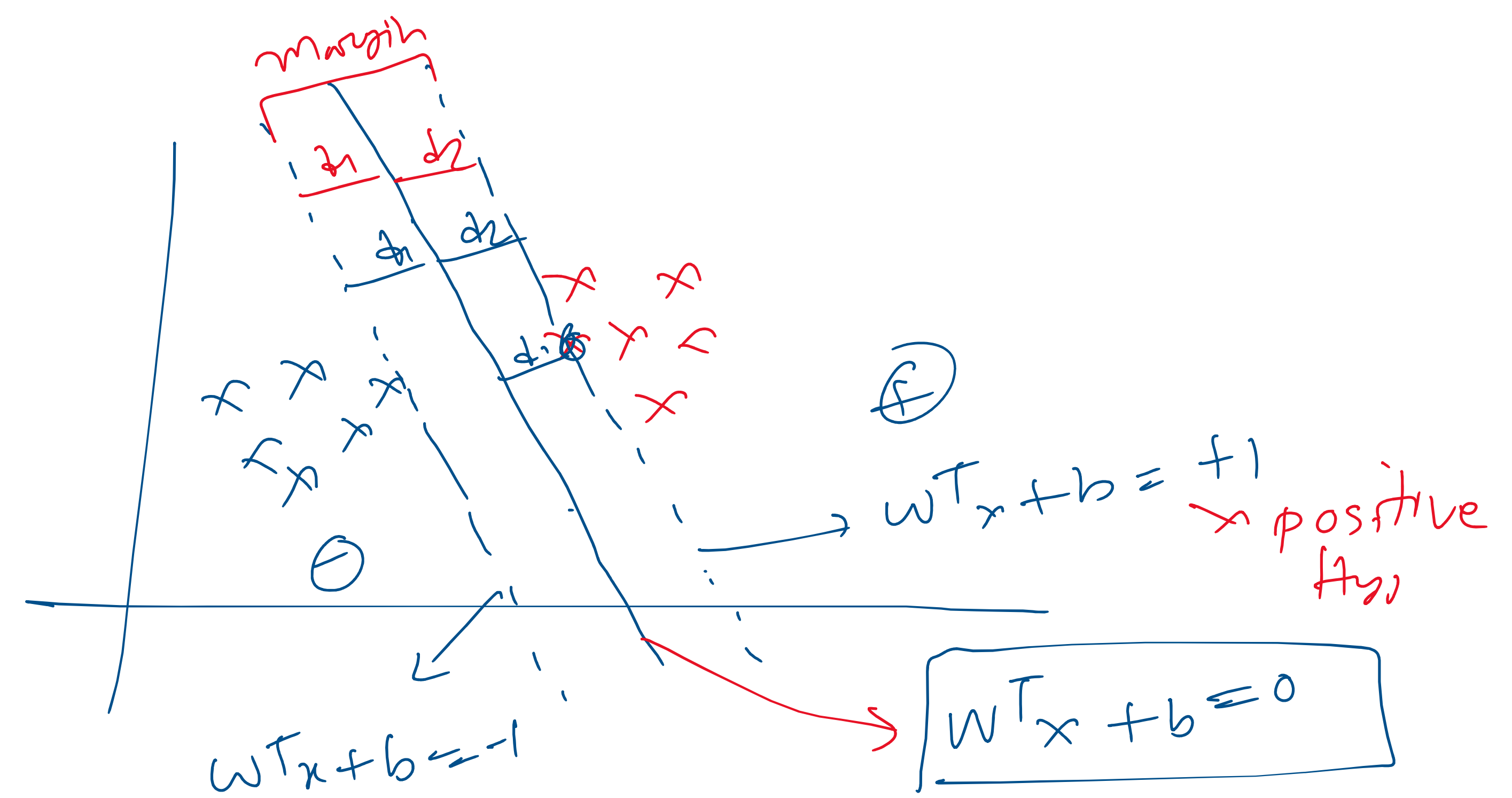
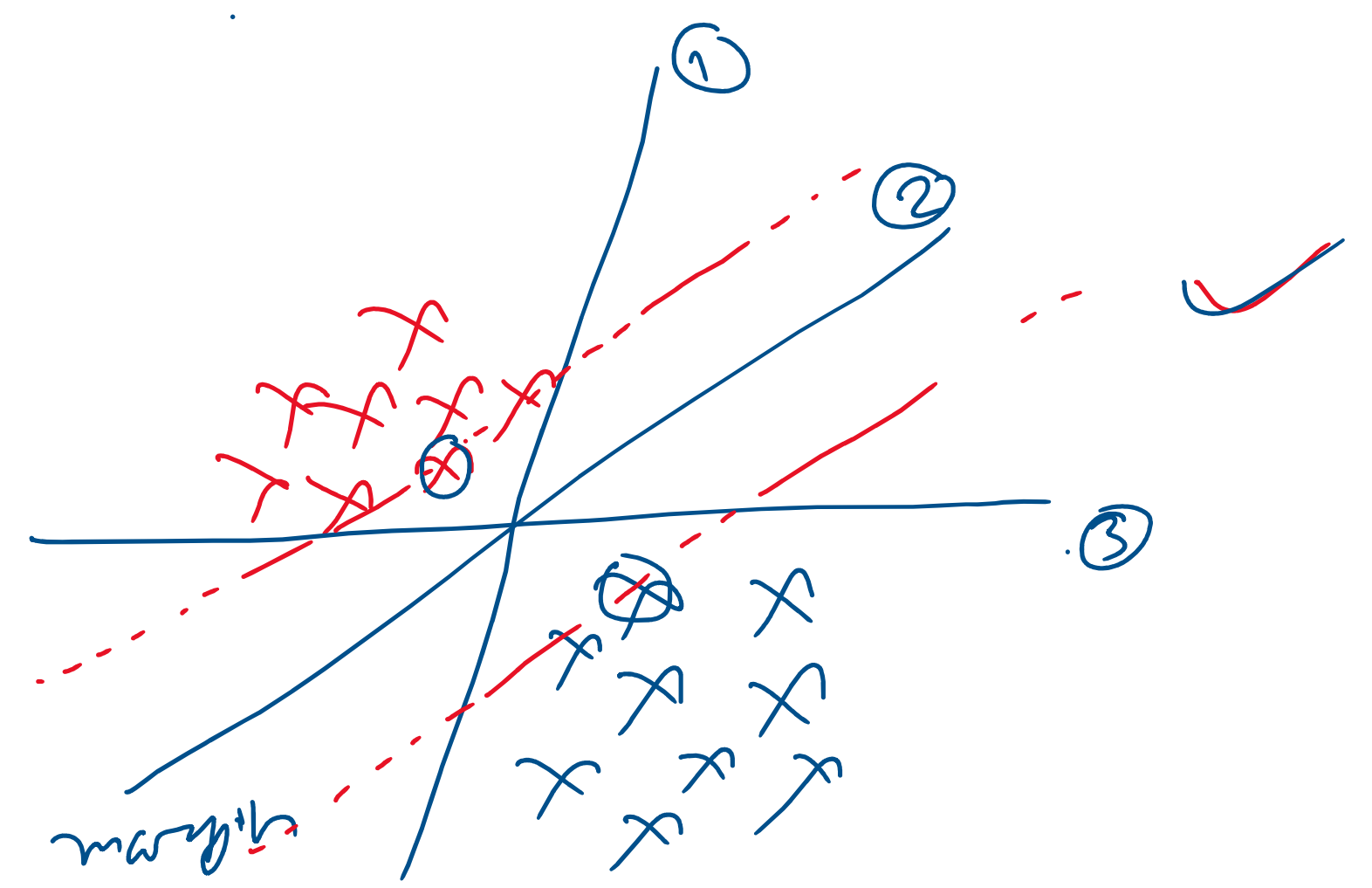
Positive hyperplane





$SVM = \text{margin} \uparrow$





negative

$$\boxed{+1}$$

$$w^T x + b \geq +1$$

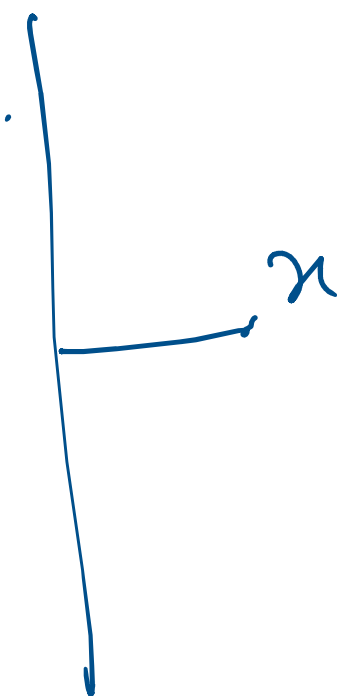
$$-1$$

$$w^T x + b \leq -1$$

$$\text{margin} = d_1 + d_2 \quad [d_1 = d_2]$$

$$= d_1 + d_1$$

$$= 2 d_1 = \frac{2}{\|w\|}$$



$$\text{Distance} = \frac{w^T x + b}{\|w\|} \rightarrow +1$$

$$= \frac{1}{\|w\|}$$

$$\text{margin} = \frac{2}{\|w\|} \quad \boxed{\uparrow}$$

$$\max f(x) = \frac{2}{\|w\|}$$

$$2x + 4y + 5$$

↖ (3,4)

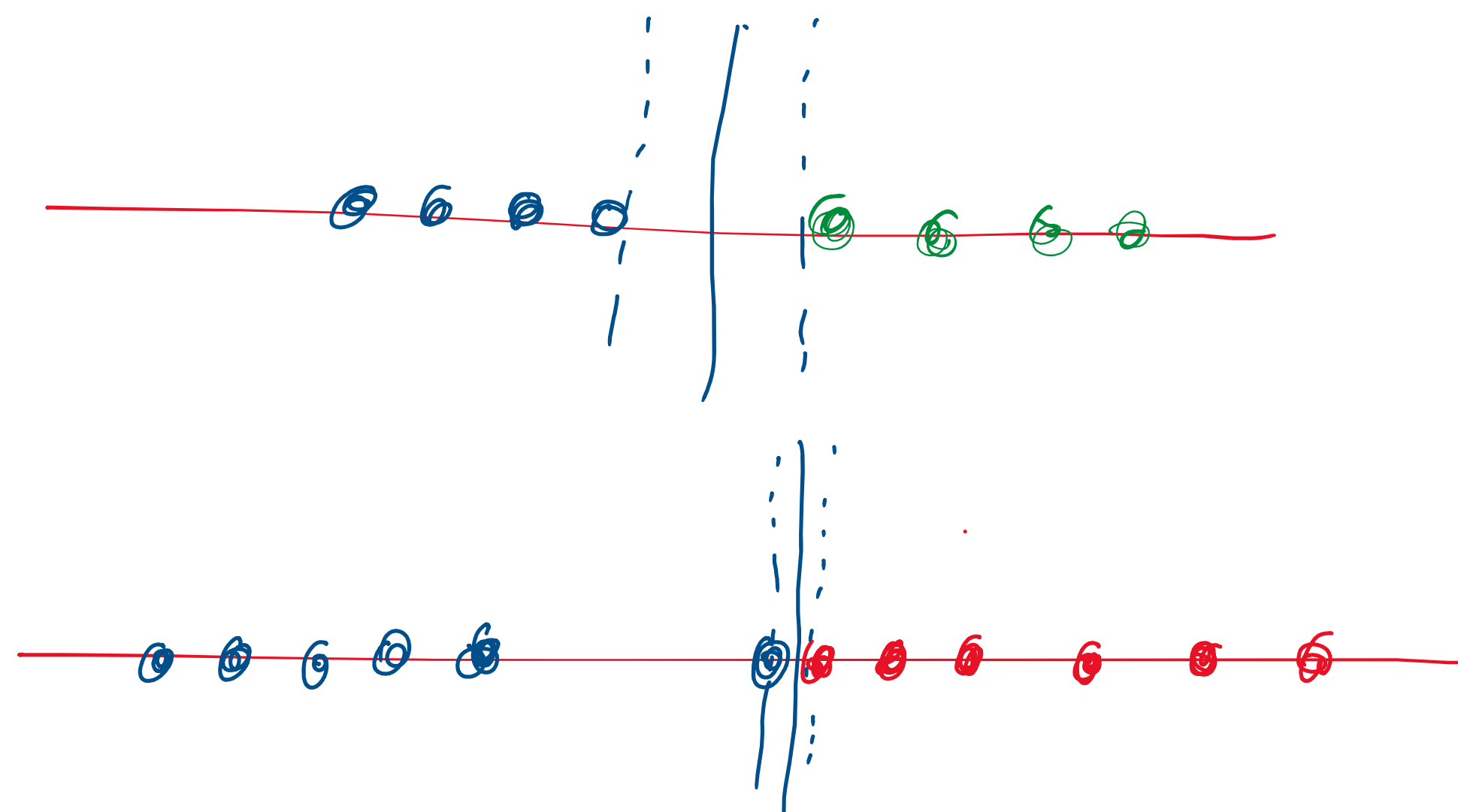
$$2 \times 3 + 4 \times 4 + 5 > 0$$

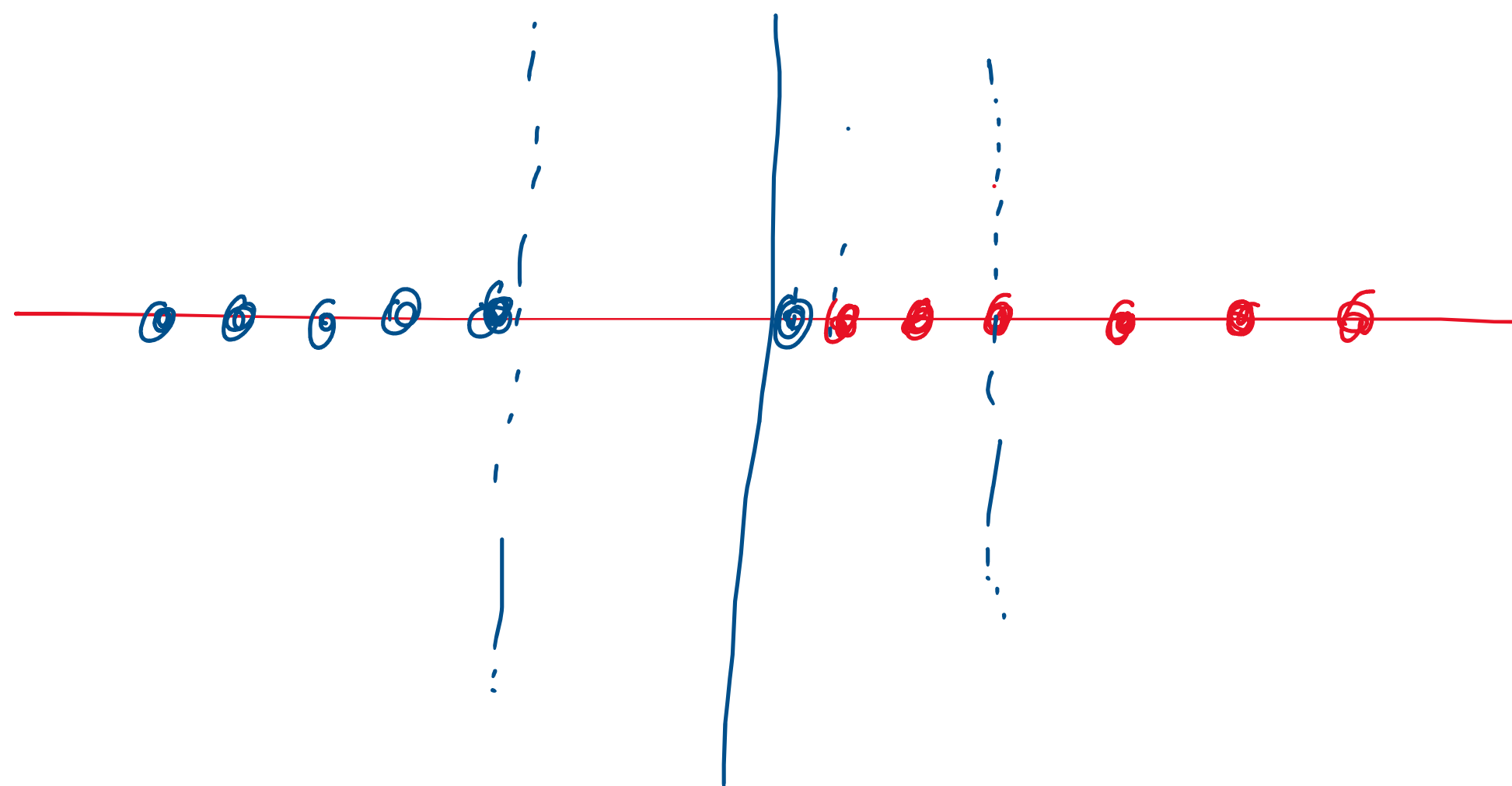
$$=$$

svm cost function

$$\min f(x) = \frac{\|w\|}{2}$$

↓
margin





$$\xi = 0 \quad \checkmark$$

margin

$$0 < \xi < 1$$

svm cost function

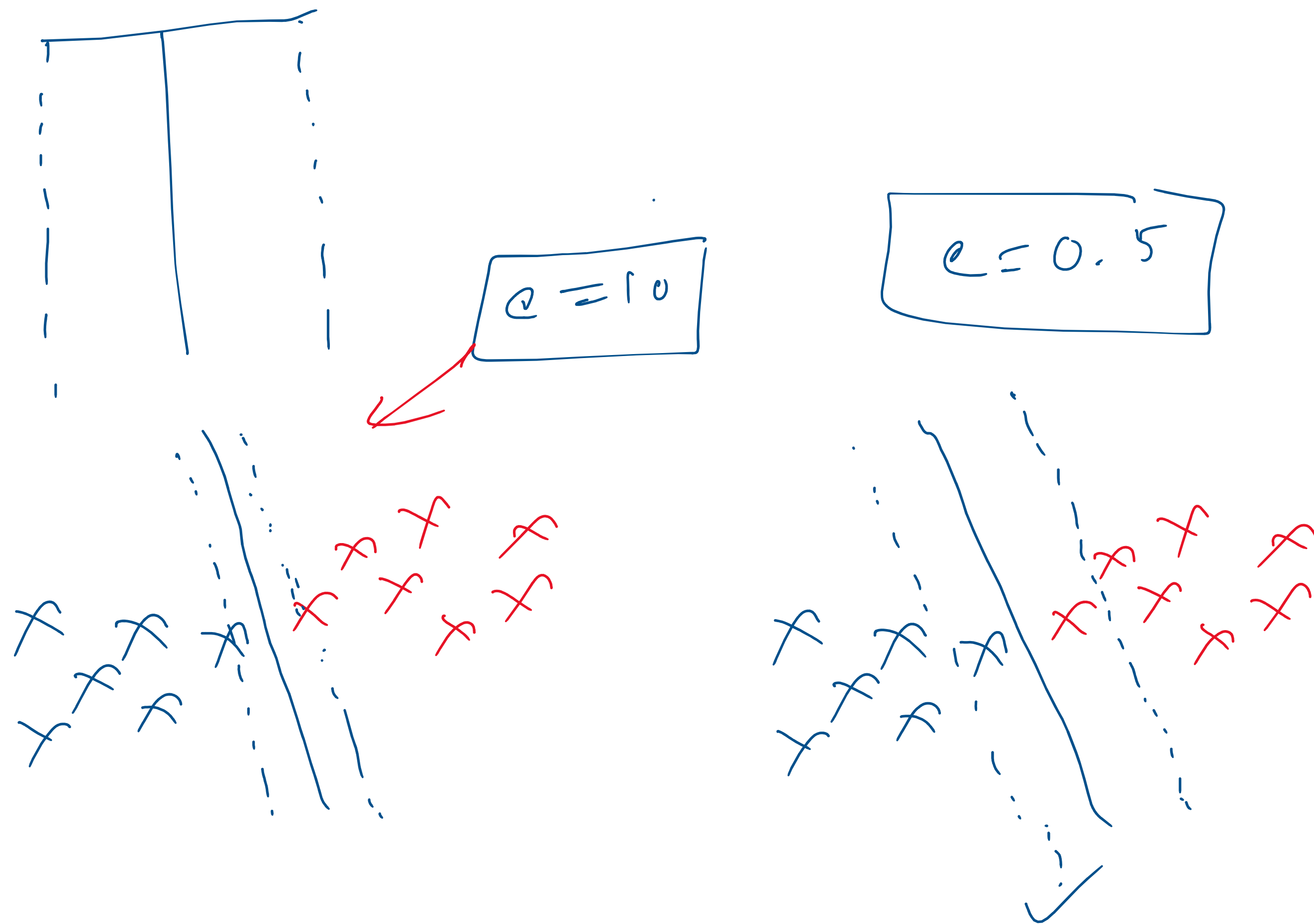
$$\min f(w) = \frac{\|w\|}{2} + \sum_{i=1}^n \xi_i$$

$\downarrow$ margin error $\downarrow$ Classification Error

$$\frac{\|w\|}{2} + c \cdot \sum_{i=1}^n \xi_i$$

$$\text{min fn} = \frac{\sum_{i=1}^n \ell_i}{2} + C \sum_{i=1}^n \ell_i$$

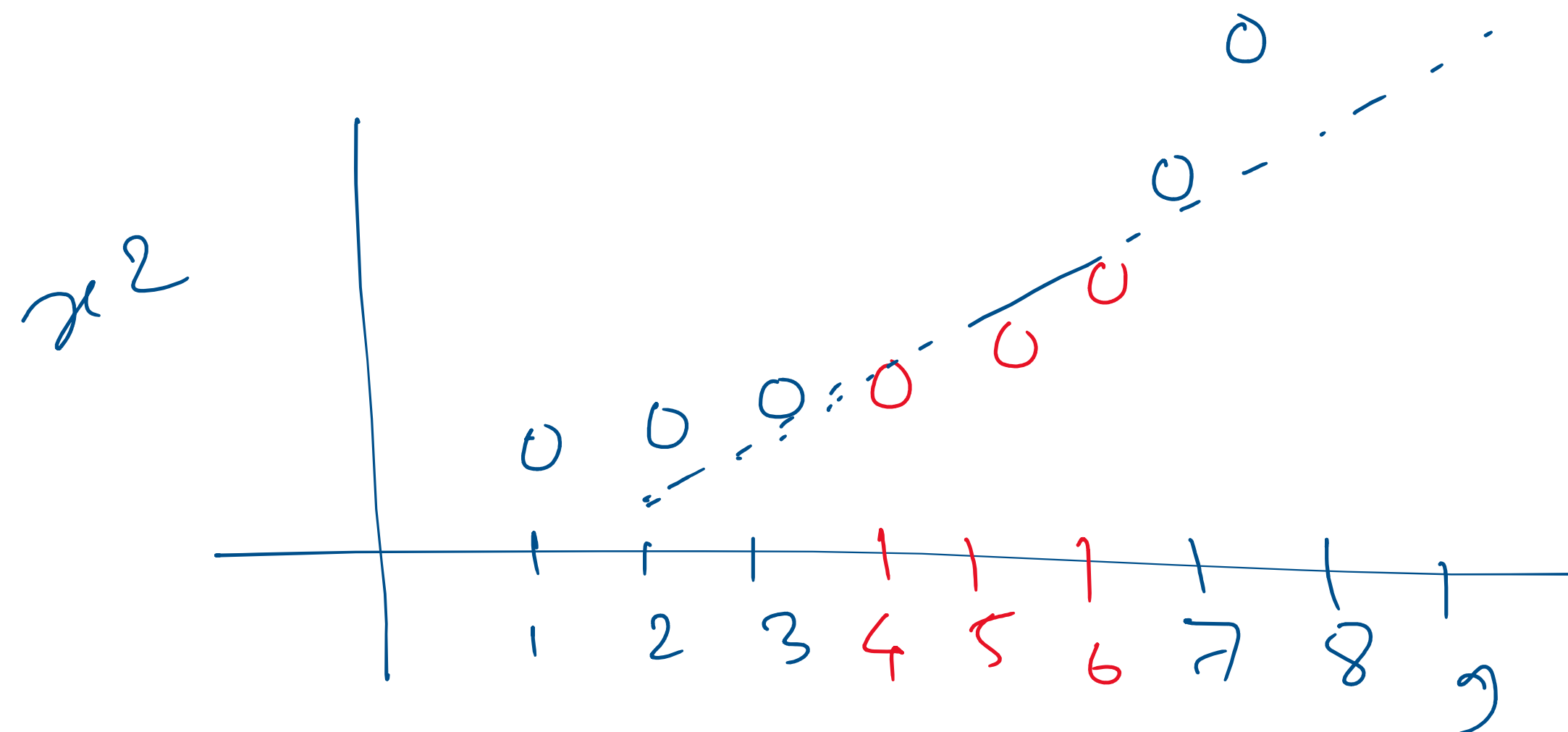
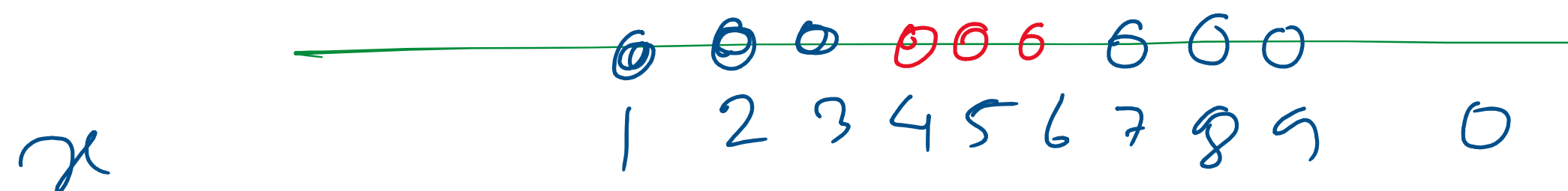
$\downarrow$ Margin Error $\downarrow$ Classification



$C \gg 1$ = misclassification Allow + Margin

0-5 = misclassification error + Margin (error)

Non-Linear



e^{-x^2}

